

Hardest games ever

Here's a list of games regarded as the most difficult games of all time by veterans. If you are a 90s kid, you'll definitely relate. <http://dgit.in/25Games>



Stalking business

Researching someone or stalking them are really the same but one can be monetised and the other cannot. How? Find out <http://dgit.in/StalkTool>

Geek life



In its short existence, Mantle did get the support of some AAA titles like Battlefield 4

hence the benefits of the low level API, like Native Multi GPU support, were still not felt beyond Windows machines using AMD's hardware.

Mantle was eventually shelved and AMD had officially advised developers to move out of the ecosystem. What it had achieved in its lifetime is a propulsion to move from DirectX and OpenGL to a much better Graphics API solution.

Why Vulkan?

With the widespread boom of Smartphones and rise of Smartphone gaming, there was a high demand for a lower overhead API that could be ideal for portable devices. But as we saw in the previous section, most efforts were either ecosystem-bound or relying on API technology developed almost 25 years ago. There was even a significant gap to be filled for the Operating Systems that did not fall within the proprietary ones that we mentioned in the earlier section.

The Khronos Group kickstarted the development of a next-gen Graphics API in July 2014 and officially announced the project with a call for participation in Dec 2014 at the SIGGRAPH. AMD handed over their Mantle API to form the base of the next gen iteration of OpenGL from Khronos. Although they initially targeted a 2015 release, the production was delayed and the Vulkan API was officially released with specifications and SDK on February 16, 2016. As of now, Android, Windows and Linux have Vulkan SDKs available.

One of the major gains with this API will be improved visuals on Linux as well as SteamOS gaming. Even Macs will gain visual improvement because Vulkan is much better performing than Metal as a low level Graphics API. A significant concern regarding SteamOS had been the lack of optimal performance in some games, which will definitely be improved with the inclusion of Vulkan support on SteamOS. Along with the specs, Khronos has also released a set of conformance tests that each manufacturer can use to demonstrate spec compliance. On the very day when the Vulkan specs were released, Nvidia, Intel (Linux), Qualcomm (Android) and Imagination technologies (also Linux) have posted their successful results of the same tests.

Note: Both the tests and the specs are available on GitHub and the Khronos group is inviting contributions highlighting inconsistencies of different drivers so that their developers can realign the drivers.

The basic nature of an API that works at a level that is lower than the existing ones is that much of the work that was the responsibility of the drivers is now to be handled by the game engines. The advantage of this is that the game engine has significantly higher information about that exact work and can make calls that the driver itself doesn't have the capability to do. And that allows the API to run with a much smaller, faster driver with support for features like multi-threading. For example, the multithreading support itself will directly



The Galaxy S7 supports Vulkan API

benefit cases where the CPU has to handle a lot of load and leaves the GPU relatively unoccupied. And any application that cannot afford performance hitches can now directly include the GPU memory management into their implementation.

As we have already emphasised, Vulkan is platform independent and already supports a number of platforms. Although it is a bare bones API, Vulkan does support the addition of layers on the stack that can deal with validations and debug data. The original intent of the Khronos group is that a large number of these layers should be created by the developers. For example, A particular developer like Intel can verify parameters pertaining to their chipset to avoid functions that are known to perform bad on their devices.

To sum it up...

A graphics API is one of those pieces of technology that we don't really bother with when setting up the latest game or preparing ourselves for our next major video hardware purchase. And thanks to developments like Vulkan, that decision has now become even safer with a universally applicable API. Combining the universal appeal of the OpenGL and the dedicated power and capability of DirectX and bringing all of that into, amongst other platforms, your smartphone, Vulkan API support is something that you should be looking forward to seeing in a lot of upcoming phones this year. **[4]**